

Exploratory Data Analysis on the Automobile Dataset

Report

Introduction

The Automobile dataset contains information about 205 vehicles from different manufacturers. It includes 26 variables describing the vehicles' physical characteristics, engine specifications, fuel economy, body style, horsepower, and price.

The objective of this exploratory data analysis was to clean and examine the dataset, identify relationships between the different variables, and answer questions about automobile prices, fuel efficiency, engine capacity, and manufacturer representation.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to clean, analyze, and visualize the data.

Data Cleaning

The dataset was first loaded into a Pandas DataFrame.

The original dataset contained:

- 205 rows
- 26 columns

The symboling and normalized-losses columns were removed because they were not required for the analysis.

Duplicate rows were then checked using the `duplicated()` method. No duplicate rows were found in the dataset.

The numerical columns horsepower, peak-rpm, and price were converted to the NumPy int64 data type after the missing values had been removed.

After the cleaning process, the final dataset contained:

- 193 rows
- 24 original analysis columns
- No duplicate rows
- No missing values

Missing Data

In the original dataset, missing values were represented by question marks (?).

When the dataset was loaded, these question marks were converted into Pandas missing values using:

```
na_values="?"
```

Missing values were found in columns such as:

- normalized-losses
- num-of-doors
- bore
- stroke
- horsepower
- peak-rpm
- price

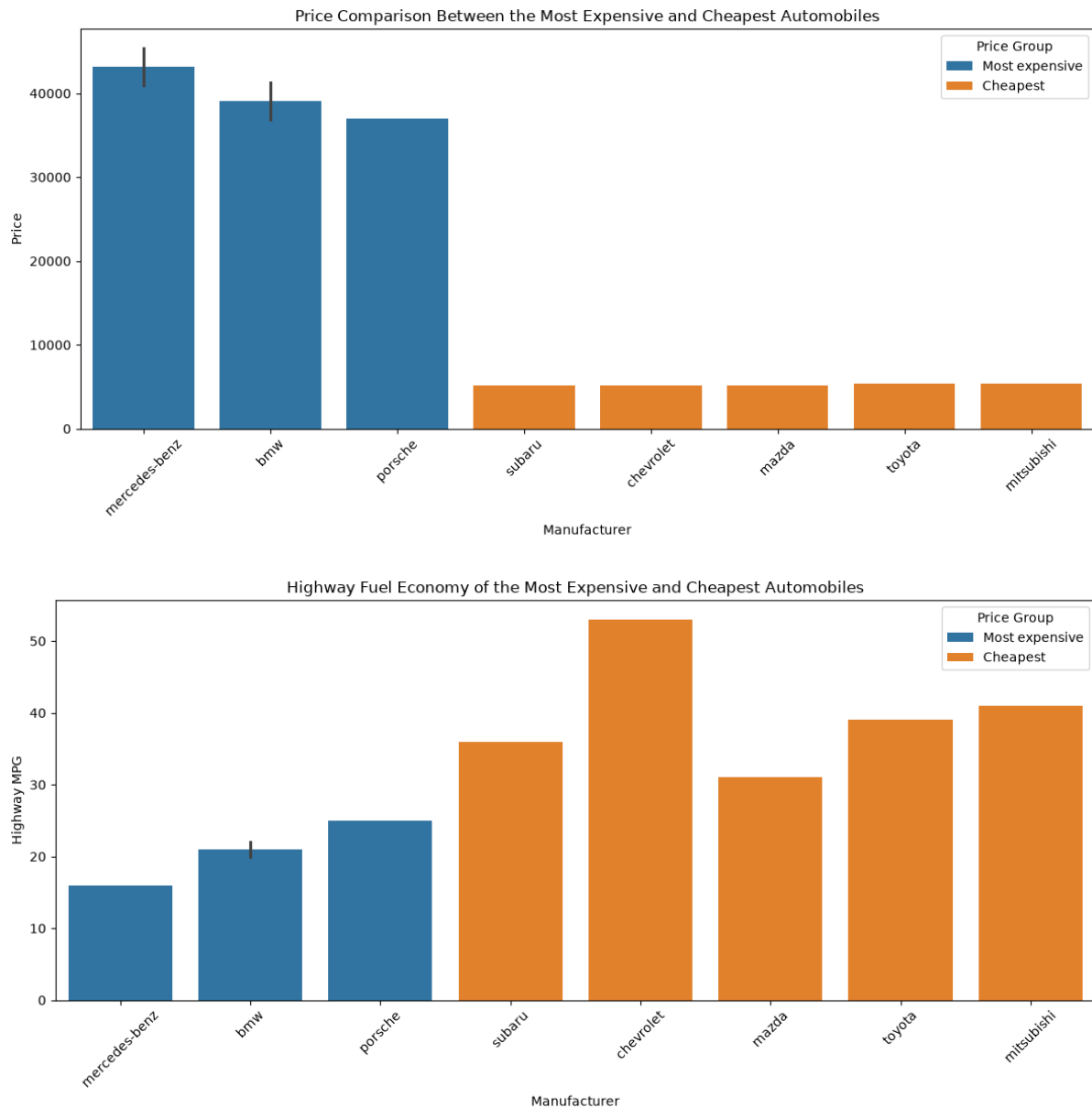
The project instructions required all entries containing missing values to be discarded. Therefore, the `dropna()` method was used to remove every row containing at least one missing value.

After this operation, 193 complete vehicle records remained available for analysis.

Data Stories and Visualizations

1. Most Expensive and Cheapest Automobiles

Visualization



Investigation

The five most expensive automobiles in the cleaned dataset were manufactured by Mercedes-Benz, BMW, and Porsche.

The most expensive vehicle was a Mercedes-Benz hardtop priced at \$45,400. It had an engine size of 304 and 184 horsepower.

The five cheapest vehicles were hatchbacks manufactured by Subaru, Chevrolet, Mazda, Toyota, and Mitsubishi. Their prices ranged from \$5,118 to \$5,389.

Finding

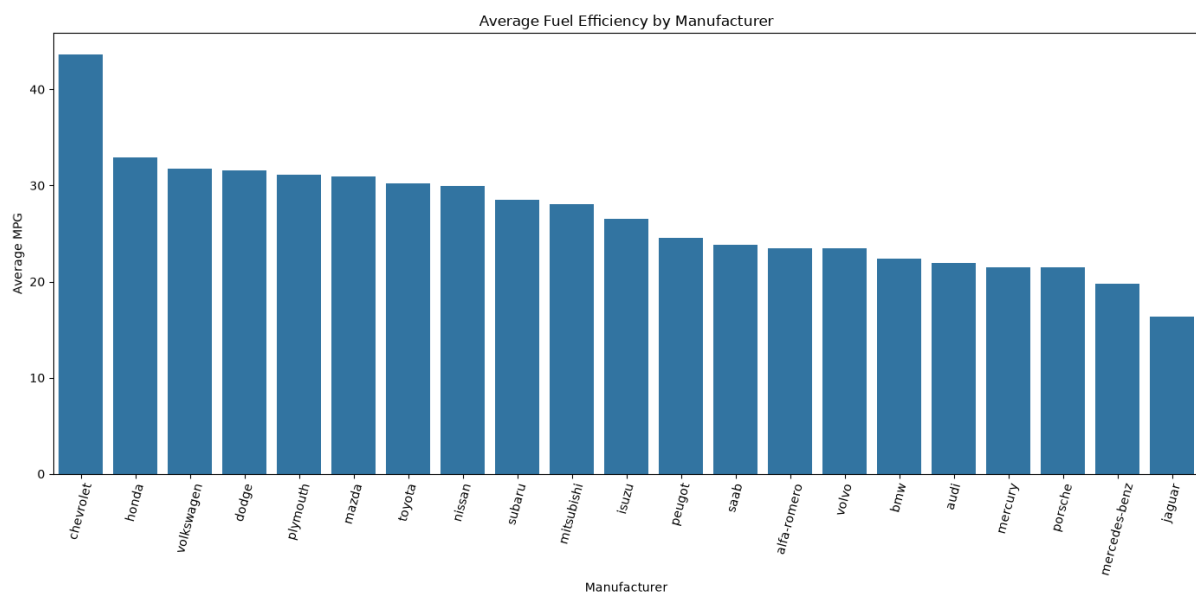
The most expensive vehicles generally had larger engines and higher horsepower than the cheapest vehicles. However, they also had considerably lower fuel economy.

For example, the most expensive vehicles achieved approximately 14–17 MPG in the city, while some of the cheapest vehicles achieved more than 30 MPG.

This indicates that a higher automobile price is associated with greater engine performance and luxury, but not necessarily with better fuel efficiency.

2. Fuel Efficiency by Manufacturer

Visualization



Investigation

Average fuel efficiency was calculated by combining city and highway MPG:

$$\text{Average MPG} = (\text{City MPG} + \text{Highway MPG}) \div 2$$

The results were then grouped by manufacturer.

Chevrolet had the highest average fuel efficiency in the cleaned dataset, with an average of approximately 43.67 MPG.

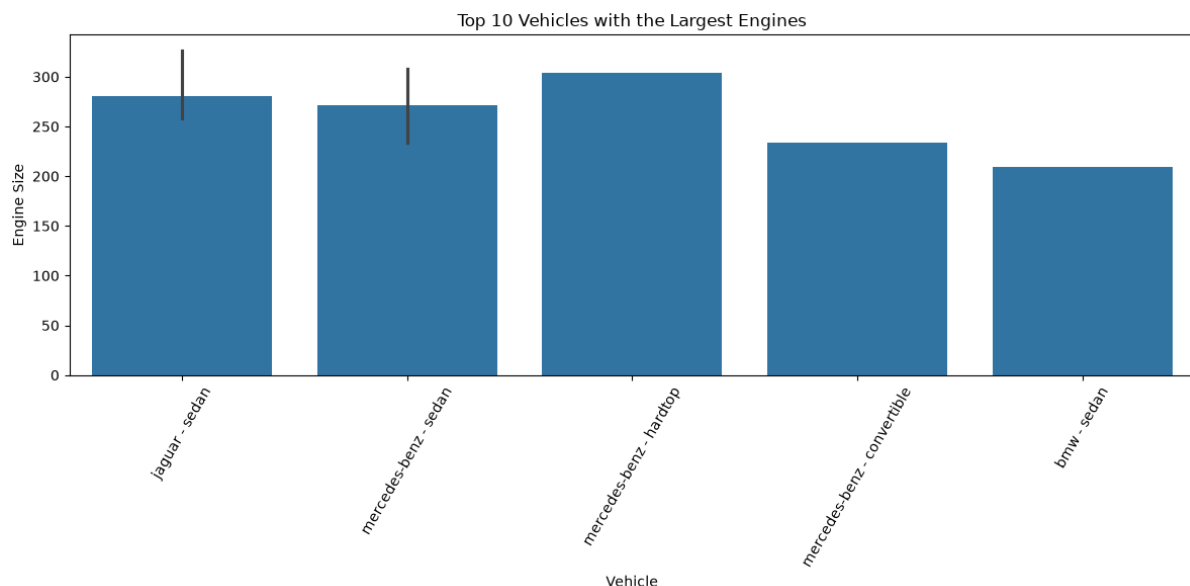
Honda had the second-highest average at approximately 32.92 MPG, followed by Volkswagen at approximately 31.75 MPG.

Finding

Chevrolet was the most fuel-efficient manufacturer represented in this dataset. However, Chevrolet had only a small number of vehicles in the dataset. Therefore, this result represents the vehicles included in this dataset and should not be interpreted as a complete comparison of every model produced by each manufacturer.

3. Automobiles with the Largest Engine Capacity

Visualization



Investigation

The vehicles were sorted according to the engine-size column.

The vehicle with the largest engine was a Jaguar sedan with an engine size of **326** and 262 horsepower.

Other vehicles with large engines were produced mainly by:

- Jaguar
- Mercedes-Benz
- BMW

The second-largest engine was found in a Mercedes-Benz sedan with an engine size of 308.

Finding

Vehicles with the largest engines were generally luxury or high-performance automobiles.

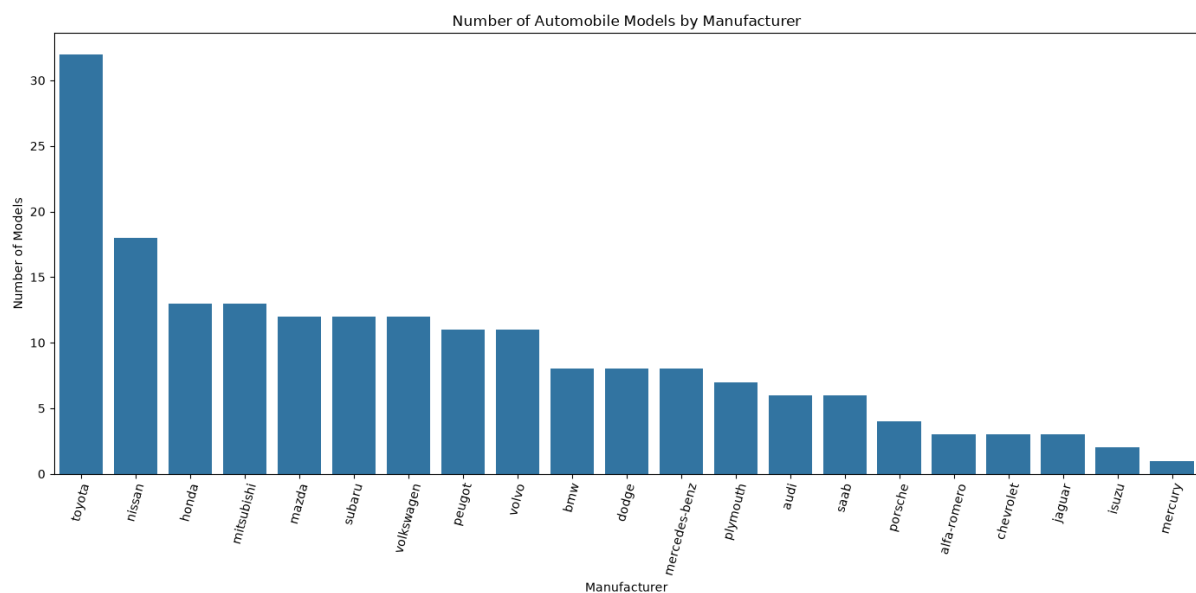
They also tended to have:

- Higher horsepower
- Higher prices
- Lower fuel economy

This suggests that engine capacity has a positive relationship with horsepower and price, but a negative relationship with fuel efficiency.

4. Manufacturer with the Most Models

Visualization



Investigation

The number of vehicle records for each manufacturer was calculated using the `value_counts()` method.

Toyota had the highest number of automobile entries in the cleaned dataset, with **32 vehicles**.

It was followed by:

- Nissan: 18 vehicles
- Mitsubishi: 13 vehicles
- Honda: 13 vehicles
- Subaru: 12 vehicles

Finding

Toyota was the most represented manufacturer in the dataset.

This means that the dataset contains more Toyota vehicle records than records from any other manufacturer. However, it does not necessarily mean that Toyota has the largest number of automobile models in the real-world market.

Overall Findings

The exploratory analysis produced several important findings.

The most expensive automobiles were mainly luxury and performance vehicles manufactured by Mercedes-Benz, BMW, and Porsche. These vehicles generally had larger engines and higher horsepower, but lower fuel efficiency.

Cheaper automobiles were generally smaller hatchbacks with lower horsepower and much better fuel economy.

Chevrolet recorded the highest average fuel efficiency among the manufacturers represented in the cleaned dataset.

Jaguar produced the vehicle with the largest engine, while Toyota was the manufacturer with the highest number of vehicle entries.

Overall, the analysis suggests that vehicle price, engine size, and horsepower are positively related. In contrast, larger engines and greater horsepower are generally associated with lower fuel efficiency.

Conclusion

The Automobile dataset was successfully loaded, cleaned, sanitised, and explored.

Unnecessary columns were removed, duplicate rows were checked, records containing missing values were discarded, and selected numerical columns were converted to the correct data types.

The visualizations made it possible to compare automobile prices, fuel efficiency, engine capacity, and manufacturer representation.

The analysis showed that expensive vehicles generally offer greater engine performance, while cheaper vehicles tend to provide better fuel economy. It also demonstrated that visualization is an effective method for identifying patterns and communicating important findings within a dataset.

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